

Heart Disease Analysis Case Study Report

1. Executive Summary

Heart disease remains one of the leading causes of mortality worldwide. Early detection and risk assessment play a crucial role in reducing complications and improving patient outcomes. This project analyzes a heart disease dataset containing demographic information and clinical measurements of patients. The objective is to identify significant risk factors, understand patterns associated with heart disease, and develop insights that can support healthcare professionals in making informed decisions.

The project includes data cleaning, exploratory data analysis, statistical analysis, visualization, and predictive modeling using Logistic Regression. The findings highlight key variables associated with heart disease and provide recommendations for healthcare organizations to improve preventive care and resource allocation.

2. Problem Statement

Healthcare organizations generate large amounts of patient data, but transforming this data into actionable insights remains a challenge.

The primary problem addressed in this study is:

"How can patient demographic and clinical data be used to identify risk factors and predict the likelihood of heart disease?"

The analysis aims to:

- Understand patient characteristics.
 - Identify significant heart disease predictors.
 - Compare patients with and without heart disease.
 - Support early diagnosis and preventive healthcare initiatives.
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3. Business Objective

The healthcare industry seeks to reduce mortality rates and treatment costs associated with cardiovascular diseases.

The key business objectives are:

- Identify major heart disease risk factors.
 - Improve patient risk assessment.
 - Support data-driven clinical decision making.
 - Optimize healthcare resource allocation.
 - Enhance preventive healthcare strategies.
 - Develop predictive models for early detection.
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4. Dataset Overview

The dataset contains patient information related to cardiovascular health.

Dataset Features

Feature Description

age	Age of patient
sex	Gender
cp	Chest pain type
trestbps	Resting blood pressure
chol	Cholesterol level
fbs	Fasting blood sugar
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise induced angina
oldpeak	ST depression
slope	Slope of ST segment
ca	Number of major vessels
thal	Thalassemia type
target	Heart disease presence

Target Variable

- 0 = No Heart Disease
 - 1 = Heart Disease
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5. Data Cleaning and Preparation

Before analysis, the dataset was cleaned and prepared.

Steps Performed

- Checked missing values.
- Verified data types.
- Removed inconsistencies.
- Created age group categories.
- Converted categorical values into meaningful labels.
- Verified duplicate records.
- Prepared data for visualization and modeling.

Outcome

The cleaned dataset was ready for exploratory analysis and predictive modeling.

6. Exploratory Data Analysis (EDA)

EDA was divided into three levels:

Basic Analysis

Key analyses included:

- Gender distribution
- Average age
- Chest pain distribution
- Blood pressure analysis
- Cholesterol analysis
- ECG distribution
- Major vessel distribution

Mid-Level Analysis

Key analyses included:

- Age vs cholesterol relationship
- Blood pressure by gender
- Exercise-induced angina analysis
- Fasting blood sugar impact
- Thalassemia analysis
- Risk factor combinations

Advanced Analysis

Key analyses included:

- Combined risk factor effects
 - Correlation analysis
 - Logistic Regression model
 - Feature importance analysis
 - Clinical measurement comparisons
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7. Key Findings

Demographic Insights

- Most patients belong to the age group of 50–60 years.
- Male patients form a larger portion of the dataset.
- Heart disease prevalence is significant among middle-aged and older adults.

Cholesterol Analysis

- Average cholesterol level: **246.50 mg/dL**
- Elevated cholesterol is common among patients.
- Cholesterol shows a mild positive relationship with age.

Chest Pain Analysis

- Typical Angina is the most frequently observed chest pain type.
- Chest pain type exhibits the strongest positive relationship with heart disease.

Heart Rate Analysis

- Patients with heart disease generally show higher maximum heart rates.
- Exercise-induced angina is associated with lower maximum heart rates.

Major Vessel Analysis

- Patients with fewer visible major vessels showed a higher incidence of heart disease.
 - Vessel count is a significant predictor of cardiovascular risk.
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8. Correlation Analysis

The strongest positive correlations with heart disease were:

Feature	Correlation
Chest Pain Type (cp)	0.432
Maximum Heart Rate (thalach)	0.420
Slope	0.344

The strongest negative correlations were:

Feature	Correlation
Exercise-Induced Angina (exang)	-0.436
Oldpeak	-0.429
Major Vessels (ca)	-0.409

Interpretation

These variables are the most influential indicators associated with heart disease status within the dataset.

9. Predictive Modeling

Logistic Regression

A Logistic Regression model was developed using all available features.

Model Performance

Metric	Value
Accuracy	81.97%
Precision	82%
Recall	85%
F1 Score	84%

Most Important Features

Positive Predictors:

- Chest Pain Type
- Slope
- Resting ECG

Negative Predictors:

- Sex
- Exercise-Induced Angina
- Thalassemia
- Major Vessels

Model Significance

The model demonstrates that clinical measurements can effectively support heart disease prediction and patient risk assessment.

10. Business Impact

Early Detection

Healthcare providers can identify high-risk patients earlier and initiate preventive measures.

Personalized Treatment

Clinical indicators can be used to design personalized treatment plans based on patient risk profiles.

Resource Allocation

Hospitals can prioritize screening programs and specialist consultations for patients exhibiting high-risk characteristics.

Cost Reduction

Preventive interventions can reduce hospitalization rates and long-term treatment costs.

Improved Patient Outcomes

Data-driven decision-making improves diagnosis accuracy and treatment effectiveness.

11. Recommendations

Based on the analysis:

1. Implement routine cholesterol screening programs.
 2. Increase cardiovascular monitoring for older patients.
 3. Prioritize patients with high-risk chest pain profiles.
 4. Develop predictive risk assessment tools using machine learning models.
 5. Encourage preventive healthcare programs targeting major risk factors.
 6. Conduct regular follow-up assessments for patients with abnormal ECG findings.
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12. Conclusion

This case study successfully analyzed heart disease patient data to identify important risk factors and clinical indicators associated with cardiovascular disease. The exploratory analysis revealed meaningful relationships between heart disease and variables such as chest pain type, maximum heart rate, exercise-induced angina, thalassemia, and major vessel count.

The Logistic Regression model achieved an accuracy of approximately 82%, demonstrating the potential of data analytics in supporting healthcare decision-making. The findings can assist healthcare providers in improving patient monitoring, enhancing preventive care strategies, and optimizing resource allocation.

Overall, this project demonstrates how data analytics can contribute to better healthcare outcomes through evidence-based decision-making and predictive modeling.